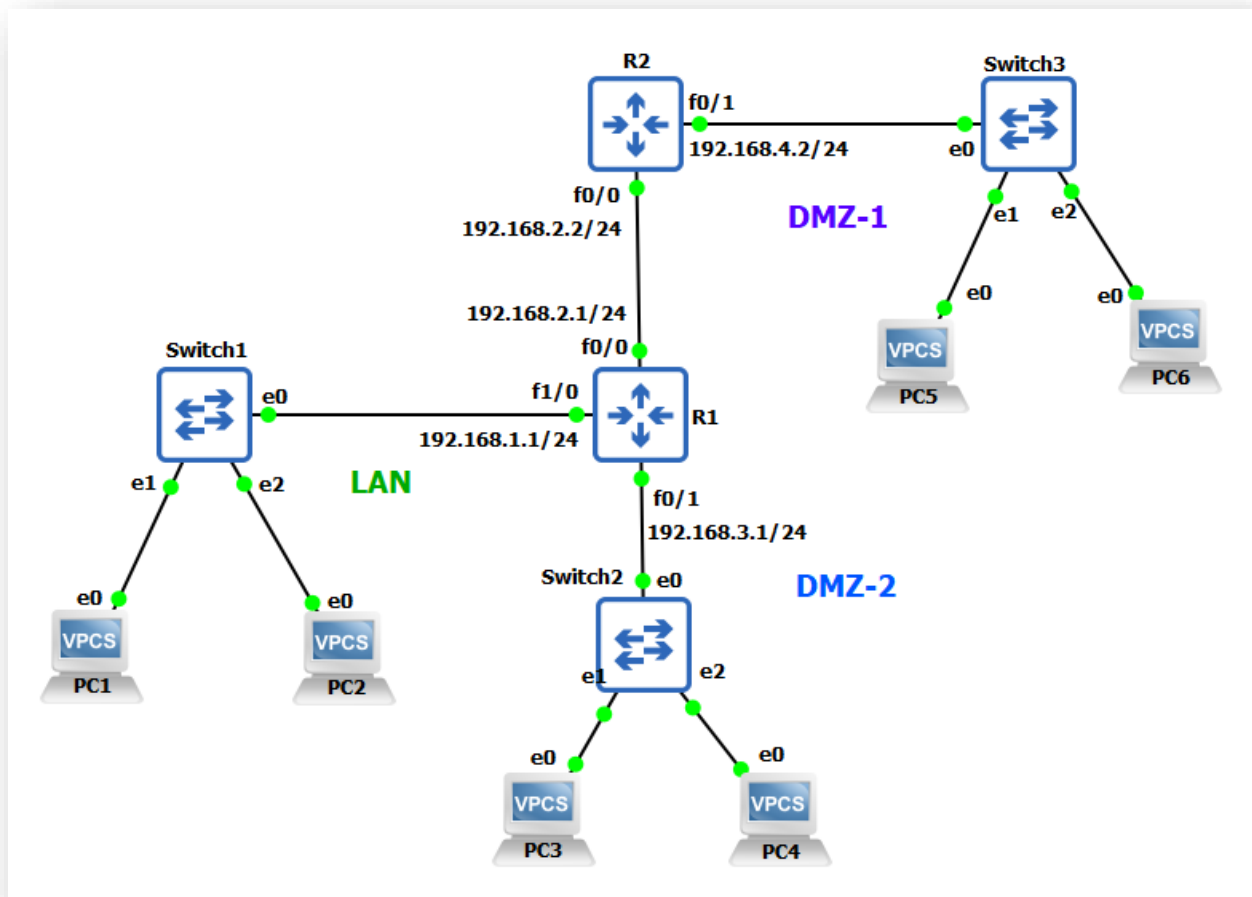


NTP & DHCP Configuration



Task

1. Configure routers with IP address as shown in topology and configure enable password as **ccna** and device access password as **cisco** (Refer Lab-1)
2. Configure R1 as NTP server and R2 as NTP client.
3. Configure MD5 authentication between NTP server and NTP client
4. Configure R1 as DHCP server to provide IP address to LAN, DMZ-1 & DMZ-2 networks and reserve first 20 IP address from each network.
5. Configure DHCP relay on R2

Task-2: Configure R1 as NTP server and R2 as NTP client.

NTP Server configuration on R1:

```

R1#config t
R1(config)#clock timezone IST +5 30
R1(config)#ntp master
R1(config)#exit
R1#clock set 21:07:30 13 Oct 2020
R1#

```

NTP Client configuration on R2:

```
R2#config t
R2(config)#clock timezone IST +5 30
R2(config)#ntp server 192.168.2.1
R2(config)#exit
R2#
```

✓ Verification & Testing:**R1#sh clock**

21:08:34.495 IST Tue Oct 13 2020

R1#

R1#sh ntp status

Clock is synchronized, stratum 8, reference is 127.127.1.1

nominal freq is 250.0000 Hz, actual freq is 250.0000 Hz, precision is 2**20
 ntp uptime is 8600 (1/100 of seconds), resolution is 4000
 reference time is E330488A.51EEC906 (21:08:50.320 IST Tue Oct 13 2020)
 clock offset is 0.0000 msec, root delay is 0.00 msec
 root dispersion is 187.77 msec, peer dispersion is 187.71 msec
 loopfilter state is 'CTRL' (Normal Controlled Loop), drift is 0.000000000 s/s
 system poll interval is 16, last update was 4 sec ago.

R1#

R2#sh clock

21:09:18.448 IST Tue Oct 13 2020

R2#

R2#sh ntp status

Clock is synchronized, stratum 9, reference is 192.168.2.1

nominal freq is 250.0000 Hz, actual freq is 250.0000 Hz, precision is 2**18
 ntp uptime is 8600 (1/100 of seconds), resolution is 4000
 reference time is E330489A.909E03EB (21:09:06.564 IST Tue Oct 13 2020)
 clock offset is -0.5821 msec, root delay is 28.45 msec
 root dispersion is 4043.10 msec, peer dispersion is 187.57 msec
 loopfilter state is 'CTRL' (Normal Controlled Loop), drift is 0.000000000 s/s
 system poll interval is 64, last update was 17 sec ago.

R2#

Note: Sometime router takes more than 4-5 minutes for clock to get synchronized. So, keep patience.

R1#sh ntp associations

address	ref clock	st	when	poll	reach	delay	offset	disp
*~127.127.1.1	.LOCL.	7	7	16	377	0.000	0.000	0.232

* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured

R1#

R2#sh ntp associations

```
address      ref clock      st  when  poll  reach  delay  offset  disp
*~192.168.2.1 127.127.1.1   8   18   64    7   12.651 -8.954 1.215
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
R2#
```

Task-3: Configure MD5 authentication between NTP server and NTP client**R1 Configuration:**

```
R1#config t
R1(config)#ntp authentication-key 1 md5 Ratan
R1(config)#ntp trusted-key 1
R1(config)#exit
R1#
```

R2 Configuration:

```
R2#config t
R2(config)#no ntp server 192.168.2.1
R2(config)#ntp server 192.168.2.1 key 1
R2(config)#ntp authentication-key 1 md5 Ratan
R2(config)#ntp trusted-key 1
R2(config)#ntp authenticate
R2(config)#exit
R2#
```

✓ Verification & Testing:**R1#sh clock**

```
21:17:46.651 IST Tue Oct 13 2020
R1#
```

R1#sh ntp status

```
Clock is synchronized, stratum 8, reference is 127.127.1.1
nominal freq is 250.0000 Hz, actual freq is 250.0000 Hz, precision is 2**20
ntp uptime is 63200 (1/100 of seconds), resolution is 4000
reference time is E3304AAA.51EC523C (21:17:54.320 IST Tue Oct 13 2020)
clock offset is 0.0000 msec, root delay is 0.00 msec
root dispersion is 0.30 msec, peer dispersion is 0.23 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is 0.000000000 s/s
system poll interval is 16, last update was 6 sec ago.
R1#
```

R1#sh ntp associations detail

```
127.127.1.1 configured, ipv4, our_master, sane, valid, stratum 7
ref ID .LOCL., time E3304ABA.51EC2794 (21:18:10.320 IST Tue Oct 13 2020)
our mode active, peer mode passive, our poll intvl 16, peer poll intvl 16
root delay 0.00 msec, root disp 0.00, reach 377, sync dist 1.00
delay 0.00 msec, offset 0.0000 msec, dispersion 0.23, jitter 0.00 msec
precision 2**20, version 4
```

```

assoc id 42517, assoc name 127.127.1.1
assoc in packets 42, assoc out packets 42, assoc error packets 0
org time E3304ABA.51EC2794 (21:18:10.320 IST Tue Oct 13 2020)
rec time 00000000.00000000 (05:30:00.000 IST Mon Jan 1 1900)
xmt time E3304ABA.51EBF1E4 (21:18:10.320 IST Tue Oct 13 2020)
filtdelay = 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
filtoffset = 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
filtererror = 0.00 0.24 0.48 0.72 0.96 1.20 1.44 1.68
minpoll = 4, maxpoll = 4
R1#

```

R2#sh clock

```
21:17:52.913 IST Tue Oct 13 2020
```

```
R2#
```

R2#sh ntp status

```
Clock is synchronized, stratum 9, reference is 192.168.2.1
```

```

nominal freq is 250.0000 Hz, actual freq is 250.0000 Hz, precision is 2**18
ntp uptime is 4400 (1/100 of seconds), resolution is 4000
reference time is E3304A8F.8686C535 (21:17:27.525 IST Tue Oct 13 2020)
clock offset is 2.2619 msec, root delay is 39.98 msec
root dispersion is 3969.80 msec, peer dispersion is 187.55 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is 0.000000000 s/s
system poll interval is 64, last update was 41 sec ago.

```

```
R2#
```

R2#sh ntp associations detail

```

192.168.2.1 configured, ipv4, authenticated+, our_master, sane, valid, stratum 8
ref ID 127.127.1.1, time E3304ACA.51EC6BDF (21:18:26.320 IST Tue Oct 13 2020)
our mode client, peer mode server, our poll intvl 64, peer poll intvl 64
root delay 0.00 msec, root disp 0.32, reach 3, sync dist 83.21
delay 19.86 msec, offset 2.2619 msec, dispersion 63.34, jitter 9.49 msec
precision 2**20, version 4
assoc id 7181, assoc name 192.168.2.1
assoc in packets 7, assoc out packets 7, assoc error packets 0
org time 00000000.00000000 (05:30:00.000 IST Mon Jan 1 1900)
rec time E3304AD0.816E14B5 (21:18:32.505 IST Tue Oct 13 2020)
xmt time E3304AD0.816E14B5 (21:18:32.505 IST Tue Oct 13 2020)
filtdelay = 32.10 39.91 19.86 28.12 28.42 39.98 36.07 0.00
filtoffset = 12.48 -5.41 2.26 12.42 -3.50 -9.49 12.37 0.00
filtererror = 0.00 0.86 0.89 0.92 0.95 0.98 1.01 16000.0
minpoll = 6, maxpoll = 10
R2#

```

Task-4: Configure R1 as DHCP server to provide IP address to LAN, DMZ-1 & DMZ-2 networks and reserve first 20 IP address from each network.

R1 Router Configuration:

```

R1#config t
R1(config)#ip route 192.168.4.0 255.255.255.0 192.168.2.2

```

```

R1(config)#ip dhcp pool LAN
R1(dhcp-config)#network 192.168.1.0 255.255.255.0
R1(dhcp-config)#default-router 192.168.1.1
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#domain-name rsatechforum.in
R1(dhcp-config)#lease 1 0 0
R1(dhcp-config)#exit
R1(config)#ip dhcp pool DMZ-1
R1(dhcp-config)#network 192.168.4.0 255.255.255.0
R1(dhcp-config)#default-router 192.168.4.1
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#domain-name rsatechforum.in
R1(dhcp-config)#lease 1 0 0
R1(dhcp-config)#exit
R1(config)#ip dhcp pool DMZ-2
R1(dhcp-config)#network 192.168.3.0 255.255.255.0
R1(dhcp-config)#default-router 192.168.3.1
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#domain-name rsatechforum.in
R1(dhcp-config)#lease 1 0 0
R1(dhcp-config)#exit
R1(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.20
R1(config)#ip dhcp excluded-address 192.168.3.1 192.168.3.20
R1(config)#ip dhcp excluded-address 192.168.4.1 192.168.4.20
R1(config)#exit
R1#

```

Note: Network 192.168.4.0/24 is not directly connected to R1. So, we need to configure routing on R1 to reach 192.168.4.0/24 for DHCP process.

✓ Verification & Testing:

PC1> clear ip

IPv4 address/mask, gateway, DNS, and DHCP cleared

PC1> ip dhcp

DDORA IP 192.168.1.21/24 GW 192.168.1.1

PC1> sh ip

```

NAME           : PC1[1]
IP/MASK         : 192.168.1.21/24
GATEWAY        : 192.168.1.1
DNS            : 8.8.8.8
DHCP SERVER     : 192.168.1.1
DHCP LEASE     : 86312, 86400/43200/75600
DOMAIN NAME    : rsatechforum.in
MAC            : 00:50:79:66:68:00
LPORT          : 10030
RHOST:PORT     : 127.0.0.1:10031
MTU:           : 1500

```

PC2> clear ip

IPv4 address/mask, gateway, DNS, and DHCP cleared

PC2> ip dhcp

DDORA IP 192.168.1.22/24 GW 192.168.1.1

PC2> sh ip

NAME : PC2[1]
 IP/MASK : 192.168.1.22/24
 GATEWAY : 192.168.1.1
 DNS : 8.8.8.8
 DHCP SERVER : 192.168.1.1
 DHCP LEASE : 86334, 86400/43200/75600
 DOMAIN NAME : rsatechforum.in
 MAC : 00:50:79:66:68:01
 LPORT : 10032
 RHOST:PORT : 127.0.0.1:10033
 MTU: : 1500

PC3> clear ip

IPv4 address/mask, gateway, DNS, and DHCP cleared

PC3> ip dhcp

DDORA IP 192.168.3.21/24 GW 192.168.3.1

PC3> sh ip

NAME : PC3[1]
 IP/MASK : 192.168.3.21/24
 GATEWAY : 192.168.3.1
 DNS : 8.8.8.8
 DHCP SERVER : 192.168.3.1
 DHCP LEASE : 86354, 86400/43200/75600
 DOMAIN NAME : rsatechforum.in
 MAC : 00:50:79:66:68:02
 LPORT : 10034
 RHOST:PORT : 127.0.0.1:10035
 MTU: : 1500

PC4> clear ip

IPv4 address/mask, gateway, DNS, and DHCP cleared

PC4> ip dhcp

DDORA IP 192.168.3.22/24 GW 192.168.3.1

PC4> sh ip

NAME : PC4[1]
 IP/MASK : 192.168.3.22/24
 GATEWAY : 192.168.3.1

```

DNS                : 8.8.8.8
DHCP SERVER        : 192.168.3.1
DHCP LEASE         : 86361, 86400/43200/75600
DOMAIN NAME       : rsatechforum.in
MAC                : 00:50:79:66:68:03
LPORT              : 10036
RHOST:PORT        : 127.0.0.1:10037
MTU:               : 1500

```

PC5> clear ip

IPv4 address/mask, gateway, DNS, and DHCP cleared

PC5> ip dhcp

DDD

Can't find dhcp server

PC6> clear ip

IPv4 address/mask, gateway, DNS, and DHCP cleared

PC6> ip dhcp

DDD

Can't find dhcp server

R1#sh ip dhcp pool

Pool LAN :

```

Utilization mark (high/low) : 100 / 0
Subnet size (first/next)    : 0 / 0
Total addresses              : 254
Leased addresses            : 2
Pending event                : none
1 subnet is currently in the pool :
Current index                IP address range    Leased addresses
192.168.1.23                192.168.1.1 - 192.168.1.254    2

```

Pool DMZ-1 :

```

Utilization mark (high/low) : 100 / 0
Subnet size (first/next)    : 0 / 0
Total addresses              : 254
Leased addresses            : 0
Pending event                : none
1 subnet is currently in the pool :
Current index                IP address range    Leased addresses
192.168.4.1                  192.168.4.1 - 192.168.4.254    0

```

Pool DMZ-2 :

```

Utilization mark (high/low) : 100 / 0
Subnet size (first/next)    : 0 / 0
Total addresses              : 254
Leased addresses            : 2
Pending event                : none

```

1 subnet is currently in the pool :

Current index	IP address range	Leased addresses
192.168.3.23	192.168.3.1 - 192.168.3.254	2

R1#

R1#sh ip dhcp binding

Bindings from all pools not associated with VRF:

IP address	Client-ID/ Hardware address/ User name	Lease expiration	Type
192.168.1.21	0100.5079.6668.00	Oct 14 2020 10:55 PM	Automatic
192.168.1.22	0100.5079.6668.01	Oct 14 2020 10:56 PM	Automatic
192.168.3.21	0100.5079.6668.02	Oct 14 2020 10:56 PM	Automatic
192.168.3.22	0100.5079.6668.03	Oct 14 2020 10:56 PM	Automatic

R1#sh ip dhcp server statistics

Memory usage	61254
Address pools	3
Database agents	0
Automatic bindings	4
Manual bindings	0
Expired bindings	0
Malformed messages	0
Secure arp entries	0

Message	Received
BOOTREQUEST	0
DHCPDISCOVER	8
DHCPREQUEST	4
DHCPDECLINE	0
DHCPRELEASE	0
DHCPINFORM	0

Message	Sent
BOOTREPLY	0
DHCPOFFER	4
DHCPACK	4
DHCPNAK	0

R1#

Task-5: Configure DHCP relay on R2

R2 Router Configuration:

```
R2#config t
R2(config)#int fa0/1
R2(config-if)#ip helper-address 192.168.2.1
R2(config-if)#exit
R2(config)#exit
R2#
```


✓ **Verification & Testing:****PC5> ip dhcp**

DDORA IP 192.168.4.21/24 GW 192.168.4.1

PC5> sh ip

NAME : PC5[1]
 IP/MASK : 192.168.4.21/24
 GATEWAY : 192.168.4.1
 DNS : 8.8.8.8
 DHCP SERVER : 192.168.2.1
 DHCP LEASE : 86238, 86400/43200/75600
 DOMAIN NAME : rsatechforum.in
 MAC : 00:50:79:66:68:04
 LPORT : 10038
 RHOST:PORT : 127.0.0.1:10039
 MTU: : 1500

PC6> ip dhcp

DDORA IP 192.168.4.22/24 GW 192.168.4.1

PC6> sh ip

NAME : PC6[1]
 IP/MASK : 192.168.4.22/24
 GATEWAY : 192.168.4.1
 DNS : 8.8.8.8
 DHCP SERVER : 192.168.2.1
 DHCP LEASE : 86247, 86400/43200/75600
 DOMAIN NAME : rsatechforum.in
 MAC : 00:50:79:66:68:05
 LPORT : 10040
 RHOST:PORT : 127.0.0.1:10041
 MTU: : 1500

R1#sh ip dhcp pool

Pool LAN :

Utilization mark (high/low) : 100 / 0
 Subnet size (first/next) : 0 / 0
 Total addresses : 254
 Leased addresses : 2
 Pending event : none

1 subnet is currently in the pool :

Current index	IP address range	Leased addresses
192.168.1.23	192.168.1.1 - 192.168.1.254	2

Pool DMZ-1 :

Utilization mark (high/low) : 100 / 0
 Subnet size (first/next) : 0 / 0

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Total addresses : 254

Leased addresses : 2

Pending event : none

1 subnet is currently in the pool :

Current index	IP address range	Leased addresses
192.168.4.23	192.168.4.1 - 192.168.4.254	2

Pool DMZ-2 :

Utilization mark (high/low) : 100 / 0

Subnet size (first/next) : 0 / 0

Total addresses : 254

Leased addresses : 2

Pending event : none

1 subnet is currently in the pool :

Current index	IP address range	Leased addresses
192.168.3.23	192.168.3.1 - 192.168.3.254	2

R1#sh ip dhcp binding

Bindings from all pools not associated with VRF:

IP address	Client-ID/ Hardware address/ User name	Lease expiration	Type
192.168.1.21	0100.5079.6668.00	Oct 14 2020 10:55 PM	Automatic
192.168.1.22	0100.5079.6668.01	Oct 14 2020 10:56 PM	Automatic
192.168.3.21	0100.5079.6668.02	Oct 14 2020 10:56 PM	Automatic
192.168.3.22	0100.5079.6668.03	Oct 14 2020 10:56 PM	Automatic
192.168.4.21	0100.5079.6668.04	Oct 14 2020 11:05 PM	Automatic
192.168.4.22	0100.5079.6668.05	Oct 14 2020 11:06 PM	Automatic

Important Commands:

```
sh clock  
sh ntp status  
sh ntp associations  
sh ntp associations detail  
sh ip dhcp pool  
sh ip dhcp binding  
sh ip dhcp server statistics
```

Ratan